

STS EEG Screening Server Update

Global STD fix + image cascades — September 2026

Four steps on each server. Do the development server first; when its results look right, do production the same way. Everything is a file copy through your normal Remote Desktop session — no scripts required.

Each study is now processed in **two passes**: the brain panel is written first, in seconds, and the image cascade follows over the next few minutes. The panel reaches the practitioner straight away, and a cascade failure can no longer destroy it.

What this changes

- **Global STD** on page 1 of every Brain Panel currently prints 1000.00 with z 2642.71. After the update it reads about 2.5, inside the normal range. No other metric moves — verified 47 of 48 identical against this server's own output.
- **An image cascade** (<study>.imagecascade.pdf, two pages per ICA component: the component view and its brain source-localization view) is written next to every panel, a few minutes after the panel itself.
- **Brain pages without OpenGL**. They are now drawn in software, so they work on the AWS servers. The old 3D viewer needed a graphics card; CLEANEEG_NO_BRAIN is no longer needed.
- **Disk**: roughly 21 MB per study with the brain pages. Production has ~200 GB on a separate data disk: 1,000 studies use about 21 GB of it. Cascades are written next to the uploaded EDF, so this assumes the upload folder is on that disk — the watchdog's GB free figure shows which drive is in use. It warns below 10 GB.

The files

On your workstation, open this folder in Explorer:

C:\BrainPanel\clean_eeg_project 2025\deploy\server-2026-09\staged

File	Goes into	What it does
files\edftotextbynameplotproc.py	project\files\	Global STD fix
files\edftotextbycommandplotproc.py	project\files\	panel pass only
files\Montage_6.py	project\files\	cascade rendering
files\dummy_gui.py	project\files\	cascade component pages
files\Component_selector.py	project\files\	same fixes, kept in sync
files\create_report_pdf.py	project\files\	cascade screen capture
process\brain_render.py	project\process\	new — brain pages, no OpenGL
mne_data\ (whole folder)	project folder (next to module7.py)	brain surfaces, ~25 MB; merge
run_cascade.py	project folder (next to module7.py)	new — the cascade pass

tomwatchdog_serialized.py	project folder (next to module7.py)	new watchdog — one at a time
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“Project folder” is the directory containing module7.py. On the development server that is C:\app\MyCleanEEG\CleanEEGProject.

On the server

In your Remote Desktop session to the server.

1. Back up the six files being replaced

In the project’s files\ folder, make a new folder called backup-2026-09 and copy these six files into it:

```
Component_selector.py
Montage_6.py
create_report_pdf.py
dummy_gui.py
edftotextbycommandplotproc.py
edftotextbynameplotproc.py
```

2. Paste the new files in

- Copy the six .py files from staged\files\ into the project’s files\ folder. Say **Yes** to overwrite.
- Copy staged\process\brain_render.py into the project’s process\ folder.
- Copy the staged\mne_data folder into the project folder. If the project already has one, let it merge and say **Yes** to overwrite — the six surface files are identical to the ones already there.
- Copy run_cascade.py and tomwatchdog_serialized.py into the project folder itself, next to module7.py.

3. Swap the watchdog

In the console window where tomwatchdog.py is running, press **Ctrl-C**. Then, in that same window:

```
set CLEANEEG_NO_BRAIN=0
```

That makes sure the brain pages are **on**. An earlier version of these steps set CLEANEEG_NO_BRAIN=1 on the servers because the old brain pages needed OpenGL. They no longer do, so the variable must be 0 or unset.

Now start the new watchdog. **Development server:**

```
py tomwatchdog_serialized.py "c:/app/STSEEGScreening/Source/Practitioners"
```

Production server:

```
py tomwatchdog_serialized.py
```

It prints monitor dir: and interpreter: first — glance at both. The production path is built in; the development server needs its own path given.

It must run in that console window — not as a service, not from a scheduled task. The cascade captures the screen, and only that session has one.

If you ran `setx CLEANEEG_NO_BRAIN 1 /M` on this server before, undo it from an elevated prompt: `setx CLEANEEG_NO_BRAIN 0 /M`.

4. Upload one study and check the result

The panel appears within seconds; the cascade follows. The console shows:

```
PROCESSING: ... (queue depth 0, N GB free)
  PANEL in 6s rc=0
CASCADE OK: ...imagecascade.pdf
  CASCADE in 66s rc=0
  DONE in 73s ...
```

Those timings are a short sample recording. A full-length study takes longer — the panel still in seconds, the cascade in minutes. Then check the files next to the EDF:

Dev	Prod	Check
☐	☐	.icale.rep.pdf — page 1, the Global STD row reads about 2.5 , not 1000.
☐	☐	.imagecascade.pdf exists, appearing a few minutes after the panel.
☐	☐	Open the cascade and page through it: component graphs on every page, no black pages .
☐	☐	After each component page there is a brain page : shaded cortex from six sides with the source in red–yellow, plus three slices.
☐	☐	Page 1 again: the four Moment 3 rows are in the hundreds to low thousands.

If the Moment 3 rows are in the **tens of thousands**: that server has numpy 2.x. Also paste staged-optional\process\detect_artifact.py into the project's process\ folder and re-run the study. One-time check per server.

That is the whole deployment. Production is the same four steps, starting the watchdog with no argument, in the console session kept logged in from the development server.

If something is wrong

1. Copy the six files from backup-2026-09 back into files\. The new brain_render.py and mne_data can stay; nothing else uses them.
2. Ctrl-C the new watchdog.
3. Start the old one: `py tomwatchdog.py`

To keep the panel fix but stop cascades, set `CASCADE_PASS = False` near the top of tomwatchdog_serialized.py and restart it. Panels are unaffected.

Nothing in this update touches data files, the reference database, or any existing output, so rollback is a complete undo.

Worth knowing

- **Why two passes:** the cascade renders inside the same code that builds the panel, so in one pass the panel came last — and a cascade crash destroyed it, which is exactly what the OpenGL failure did on the dev server. Split in two, the panel is on disk before the cascade starts.
- **Why the watchdog changes:** the old one starts a new process for every upload with no limit. Two cascades rendering at once each capture the other's window, and the result is wrong with no error. The new one runs one study at a time, waits for an upload to finish writing, and kills anything past 25 minutes.
- **Development server only:** its screen is live only while you are connected. Stay connected while a cascade renders. Production is unaffected — its session is held by the console logged in from dev.
- **Screen saver / lock screen:** keep both off on any machine rendering cascades. A lock screen firing mid-render produces black pages.
- **Older panels:** every panel issued before this update carries the bogus Global STD row. Nothing else on them is wrong; their out-of-bounds count is one too high.
- **Longer documents:** README.md explains each change. MESA_OPENGL.md is out of date: the brain pages no longer need OpenGL.